	AzureSky	
AzurESky "ensure quality of test"	Calorific Value determination	Document no: AS007IW
Compiled by:	Sifiso Nyambi - Laboratory Manager	
Approved by:	Nonhlanhla Msibi - Operations Manager	

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1. SCOPE

This procedure describes the accepted method to be followed for the determination of calorific value of coal at constant volume and at controlled temperature that is between 19^0C to 26^0C using a bomb calorimeter.
Test results are verified by using a control sample and plotting the results onto a graph. Predetermined upper and lower limits are used to identify results that are out of control. These results, then, initiate an investigation by the analyst and lab manager to determine the actual cause.

NOTE: ONLY PERSONNEL WHO ARE DECLARED COMPETENT ARE TO PERFROM LABORATORY ACTIVITIES. PERSONNEL NOT DECLARED ARE TO STRICTLY WORK UNDER SUPERVISION OF A LAB MANAGER/SUPERVISOR OR COMPETENT PERSONNEL.

2. References: ISO 1928: 2020 (Edition 3)

3. Reagents

- Oxygen at a pressure high enough to fill a combustion bomb to 3MPa, with an assay of at least 99.5% volume fraction and free from combustible matter.
- Benzoic acid tablets (~0.50 g).

4. Apparatus

- Combustion Bombs: Made of stainless steel with brass threads and capable of withstanding safely the pressures developed during combustion.
- Chiller
- Gas filling station set at 2500 Kpa.
- Balance accurate to 0.1mg and 0.01 mg if the sample is of the order of 0.5g or less
- Crucibles made from stainless steel.

5. Sample

The sample shall be prepared according to instructions on the preparation of sample as per ASOO1IW form. The top size of the analysis for CV determination should be < 212 µm.

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6. Pre-Inspection checklist

- Check the calorimeter structure for any defects and cleanliness.
- Check the oxygen loading station that enough gas is supplied and no leaks in the system.
- No oil or lubricant has to be used on the equipment.
- Check your gas supply gauge if the set pressure does not exceed 2500Kpa.
- Inspect the bomb vessel and lid for visual damage such as cracks and inner “thread” damage.
- Inspect the lid seal O-ring for signs of perishing.
- Check the cooling assembly for any leaks, visible defects or damage.
- Inspect the electrodes assembly for visible damage and that there are no deposits or built ups on any part. If there is any built up noted, use sandpaper to gently remove it.

- Inspect the crucibles for cleanliness and should not be dented. Dented crucibles must be rejected and ejected from the system.

Note: Ensure all equipment and materials used are verified against factory specifications and logged in the equipment register prior to use. Calibrations (where necessary) and verification of equipment are to be attended prior to use. Calibration services shall only be sourced from ISO/IEC 17025 accredited labs. Verification certificates from manufacturers are accepted for new equipment provided to address traceability.(ISO 17025; Clause 6.4.)

7. Definition

Calorific Value is the measure of the energy output generated by coal sample during combustion. The CV of a substance is influenced by the temperature and an increase in temperature will increase the CV and vice-versa. Thus, it is essential for the laboratory conditions to be controlled.

8. Procedure

1. Prepare the bomb vessel by ensuring that it is fit for usage as per section 6.
2. Place crucible inside the balance, press the “tare” or “zero” key.
3. Weigh sample ensuring that the mass falls within the range 0.5000-0.5099 g.
4. Open the drycal software/application.

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5. Press F2 on the keyboard/monitor, enter the sample ID and mass weighed.
6. Take the bomb cap and place the crucible on the designated part on the ground electrode.
7. Ensure that the cotton is buried underneath the sample.
8. Take the cap and place it on the bomb, the number on the bomb must face towards you and the hole where the valve pin goes in should be at 5 (like the clock) position. Turn the bomb cap towards the 6th position (i.e. the hole of the cap should be in line with the pin. Push the pin in.
9. Fill in the oxygen until the needle is at 2500kpa on the regulator.
10. Carefully place the bomb inside the calorimetry and then close the lid.

9. Repeatability and reproducibility limit

- The repeatability limit should be <0.12 MJ/kg.
- The reproducibility limit: This refers to analysis carried out by two different people/ labs/days, the difference between these measurements should not be > 0.30 MJ/kg.

10. Calibration

NB: Calibration should only be done if the control is outside the repeatability limit, and you have exhausted all possible options that could cause this (such as, low oxygen pressure, leaks, proper fitting of all bombs components, firing wire).
If the assigned value of a CRM is 20MJ/kg and you obtain 15 MJ/kg (or any other value that is far off). This is not a problem associated with the need to calibrate; there is a problem with the workings of the instrument itself. Check the calorimeter/pressure and inspect the bomb.

- Weigh the benzoic acid tablet and then follow steps 3-10 under section 8.
- The benzoic tablet should be analysed at least 5 times.
- To calibrate: Click F8 and select at least three values from the ones you just analysed.
- The assigned value for benzoic acid tablets is ~26.4 MJ/kg (check the bottle for the exact value).
  The standard deviation should be less than 0.2, if not analyse 2-3 more pills and check again using different values. Under no circumstances should you go ahead with the calibration if standard deviation is >0.2. If you are not sure about what to do, inform the laboratory manager.
- **Use the AS003TF- CV CALIBRATION FORM to record your data.**

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11. Revision list

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